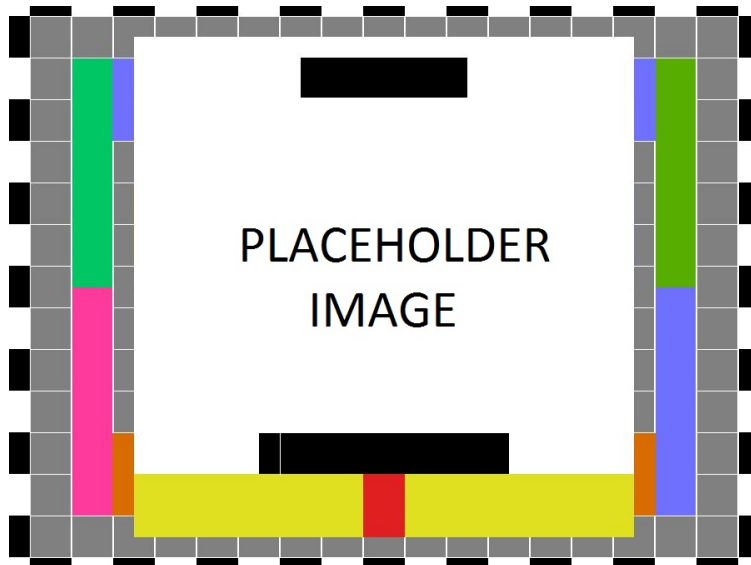


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**SYSTEM REQUIREMENTS SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



TEAM NAME
PRODUCT NAME

ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN
ADA LOVELACE
CHARLES BABBAGE

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1 INTRODUCTION

This System Requirements Specification (SRS) describes the requirements for the Product Name system. It is organized in accordance with the SRS information item defined in ISO/IEC/IEEE 29148:2018 [1], the current international standard for requirements engineering (which supersedes the withdrawn IEEE Std 830-1998). Replace this paragraph with a one-paragraph synopsis of the product and the purpose of this document.

1.1 PURPOSE

State the purpose of this SRS and identify the intended readership (customer, sponsor, team members, instructors, future maintainers). Explain that this document is the authoritative, agreed baseline of what the system shall do and shall not do, and that requirements in it may not be changed without agreement of the intended customer/sponsor.

1.2 SCOPE

Name the software/system product(s) to be produced. Explain what the product will and will not do, and describe the application and benefits of the product. Be careful to bound the scope: describe the boundary between this system and the external systems, users, and environment it interacts with.

1.3 PRODUCT OVERVIEW

Provide the context needed to understand the requirements that follow. Per ISO/IEC/IEEE 29148, address at least the four topics below.

1.3.1 PRODUCT PERSPECTIVE

Describe how this product relates to other products or systems. Is it self-contained, a component of a larger system, or a replacement for an existing system? Include a high-level context diagram showing the system boundary, its users, and the external systems/interfaces it connects to.

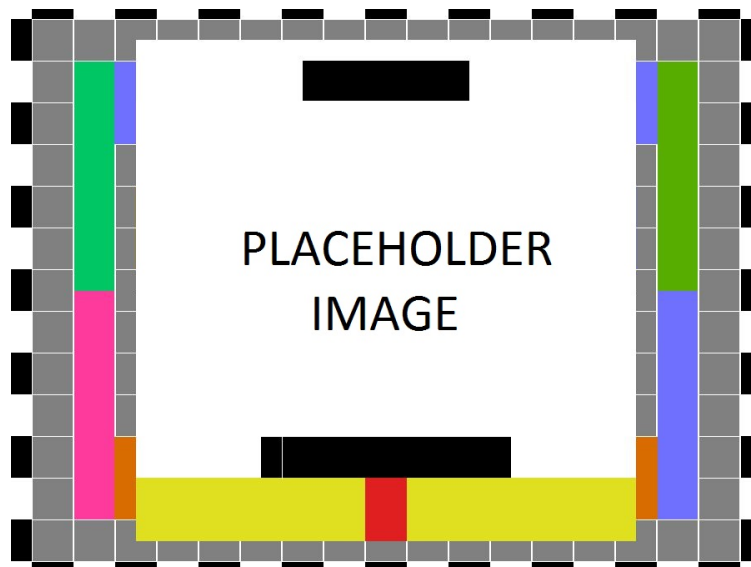


Figure 1: Product Name system context diagram

1.3.2 PRODUCT FUNCTIONS

Summarize, at a high level, the major functions the product will perform. A bulleted list or a simple functional decomposition diagram is appropriate. Detailed, individually numbered requirements are specified in Sections 3–11; this subsection is only an orientation.

1.3.3 USER CHARACTERISTICS

Identify each class of user (end user, administrator, maintainer, external system) and the characteristics that affect requirements: expertise level, frequency of use, physical or accessibility considerations, and any assumptions about training.

1.3.4 ASSUMPTIONS & DEPENDENCIES

List the factors that, if changed, would change the requirements in this document. Distinguish assumptions (believed true but not guaranteed) from dependencies (relationships on external deliverables, systems, or organizations). Keep detailed project assumptions in the Project Charter; here, list only those that directly shape requirements.

1.4 DEFINITIONS, ACRONYMS, AND ABBREVIATIONS

Define all terms, acronyms, and abbreviations required to interpret this document. A two-column table works well.

Table 2: Definitions, acronyms, and abbreviations

Term	Definition
SRS	System Requirements Specification
Term	Definition of the term as used in this document

2 REQUIREMENTS ENGINEERING APPROACH

This section defines the conventions used to write and manage the requirements in this document. Remove the explanatory paragraphs from your final draft, but retain the tables and conventions your team adopts.

2.1 REQUIREMENT IDENTIFICATION & NUMBERING

Every requirement in this document has a unique, persistent identifier so that it can be traced through design, implementation, and test. This template uses the scheme [TYPE] - [NN], where TYPE denotes the requirement category and NN is a zero-padded sequence number, e.g. FR-01 (functional), PE-03 (performance), SE-02 (security). Identifiers are never reused or renumbered once assigned; a retired requirement is marked *deprecated* rather than deleted.

Table 3: Requirement type prefixes

Prefix	Requirement type
EI	External interface
FR	Functional
US	Usability
PE	Performance
PK	Packaging & deployment
SA	Safety
SE	Security & privacy
MS	Maintenance & support
QA	Other quality attribute

2.2 REQUIREMENT ATTRIBUTES

Each requirement is specified using the attribute set below, adapted from the requirement attributes recommended in ISO/IEC/IEEE 29148:2018 [1]. Every requirement in Sections 3–11 provides each of these attributes.

- **Description** — A single, testable statement of the required capability or characteristic, written using “shall.”
- **Rationale** — Why the requirement exists and the benefit it provides; this justifies the requirement and aids future change decisions.
- **Source** — The origin of the requirement (customer, sponsor, named team member, regulation, standard, CSE Senior Design specifications, etc.).
- **Priority** — The relative importance of the requirement (see below).
- **Verification Method** — How conformance will be demonstrated (see below).
- **Constraints** — Realistic constraints relevant to the requirement (economic, environmental, social, political, ethical, health & safety, manufacturability, sustainability), as applicable.
- **Applicable Standards** — Any specific standards that govern the requirement.

2.3 PRIORITY SCHEME

- **Critical** — must have, or the product is a failure
- **High** — very important to customer acceptance and desirability
- **Moderate** — should have for proper product functionality
- **Low** — nice to have; will include if time/resources permit
- **Future** — not feasible in this version; recorded for a future release (see Section 13)

2.4 VERIFICATION METHODS

Each requirement declares how it will be verified, using one of the four standard methods. The Verification & Traceability section (Section 12) collects these into a requirements verification matrix.

- **Inspection (I)** — verification by visual examination of the item or its documentation.
- **Analysis (A)** — verification by technical evaluation, modeling, or calculation.
- **Demonstration (D)** — verification by exercising the system and observing behavior, without detailed measurement.
- **Test (T)** — verification by controlled execution against defined inputs with measured, recorded results.

2.5 CHARACTERISTICS OF WELL-FORMED REQUIREMENTS

Requirements in this document are written to satisfy the characteristics defined in ISO/IEC/IEEE 29148:2018: each requirement should be *necessary, appropriate, unambiguous, complete, singular, feasible, verifiable, correct, and conforming*, and the requirement set as a whole should be *complete, consistent, feasible, comprehensible, and able to be validated*. Avoid implementation detail unless it is a genuine constraint; specify *what* is required, not *how* it is built (design decisions belong in the Architectural and Detailed Design Specifications).

3 EXTERNAL INTERFACE REQUIREMENTS

Include a header paragraph specific to your product here. External interface requirements define the interactions between the system and everything outside its boundary: users, hardware, other software, and communications. Per ISO/IEC/IEEE 29148:2018, specify each interface's content and format precisely enough that an independent team could implement either side. Where helpful, summarize the data crossing an interface in a table (name, description, direction, units/format, range).

3.1 EI-01: [INTERFACE NAME]

3.1.1 DESCRIPTION

The system shall ... (a single testable statement describing the interface, e.g. *"The system shall expose a REST API over HTTPS accepting JSON request bodies encoded in UTF-8."*). Identify whether the interface is a user interface, hardware interface, software interface, or communications interface, and reference the relevant data elements.

Table 4: EI-01 data elements

Element	Description	Direction	Format / Units
element 1	description	in	type, range
element 2	description	out	type, range

3.1.2 RATIONALE

Why this interface is required and the benefit it provides.

3.1.3 SOURCE

Customer, sponsor, named team member, standard, etc.

3.1.4 PRIORITY

Critical / High / Moderate / Low / Future

3.1.5 VERIFICATION METHOD

Inspection / Analysis / Demonstration / Test

3.1.6 CONSTRAINTS

Any realistic constraints applicable to this interface.

3.1.7 APPLICABLE STANDARDS

e.g. IETF RFC 8259 (JSON), USB 2.0, IEEE 802.11, etc.

3.2 EI-02: [INTERFACE NAME]

Repeat the attribute block for each external interface.

4 FUNCTIONAL REQUIREMENTS

Include a header paragraph specific to your product here. Functional requirements are the directly observable features and functions of the product encountered by its users — the “look and feel” and behavior that each end user should expect the product to do and/or not do. These correspond to the customer-facing requirements agreed with the intended customer/user/sponsor, and must not be changed without their specific agreement. Each requirement below is written as a singular, verifiable “shall” statement and carries the attribute set defined in Section 2.

4.1 FR-01: [REQUIREMENT NAME]

4.1.1 DESCRIPTION

A detailed, singular, and verifiable statement of the feature/function. For example: *The GUI background shall be slate blue (#007FFF, six-digit hexadecimal), so that the GUI matches other software products offered by the customer.* It is acceptable and advisable to include drawings/graphics if they aid understanding.

4.1.2 RATIONALE

The customer need or benefit that this requirement satisfies. This justifies the requirement and supports future change decisions.

4.1.3 SOURCE

The origin of the requirement (customer, sponsor, named team member, federal regulation, CSE Senior Design project specifications, etc.).

4.1.4 PRIORITY

Critical / High / Moderate / Low / Future

4.1.5 VERIFICATION METHOD

Inspection / Analysis / Demonstration / Test — how conformance will be shown.

4.1.6 CONSTRAINTS

Realistic constraints relevant to this requirement (economic, environmental, social, political, ethical, health & safety, manufacturability, sustainability), as appropriate.

4.1.7 APPLICABLE STANDARDS

Any specific standards that apply (e.g. a color specification, a data-format RFC, an accessibility guideline). Research and document which standards will be followed.

4.2 FR-02: [REQUIREMENT NAME]

Repeat the attribute block for each functional requirement.

5 USABILITY REQUIREMENTS

Include a header paragraph specific to your product here. Usability requirements specify measurable objectives for how effectively, efficiently, and satisfactorily identified users can achieve their goals (see the quality model of ISO/IEC 25010:2023 [2]). Prefer measurable targets — learnability, task completion time, error rate, accessibility conformance — over subjective language.

5.1 US-01: [REQUIREMENT NAME]

5.1.1 DESCRIPTION

A measurable usability statement, e.g. *A first-time user shall complete the primary task within 3 minutes without written instructions in at least 8 of 10 usability trials.*

5.1.2 RATIONALE

Why this level of usability matters to the customer or end users.

5.1.3 SOURCE

Source of the requirement.

5.1.4 PRIORITY

Critical / High / Moderate / Low / Future

5.1.5 VERIFICATION METHOD

Typically Demonstration or Test (e.g. moderated usability sessions, accessibility audit).

5.1.6 CONSTRAINTS

Applicable constraints (accessibility mandates, device limitations, etc.).

5.1.7 APPLICABLE STANDARDS

e.g. WCAG 2.2 for web accessibility, Section 508, ISO 9241 ergonomics, as applicable.

5.2 US-02: [REQUIREMENT NAME]

Repeat the attribute block for each usability requirement.

6 PERFORMANCE REQUIREMENTS

Include a header paragraph specific to your product here. Performance requirements address measurable, time- or capacity-bounded behavior: how fast critical operations must complete; start-up and shut-down times; throughput and capacity; battery life; setup time; etc. Each must be quantified so it can be verified by test or analysis.

6.1 PE-01: [REQUIREMENT NAME]

6.1.1 DESCRIPTION

A quantified statement, e.g. *The system shall return search results within 500 ms (95th percentile) for a catalog of up to 100,000 items.*

6.1.2 RATIONALE

Why this level of performance is required.

6.1.3 SOURCE

Source of the requirement.

6.1.4 PRIORITY

Critical / High / Moderate / Low / Future

6.1.5 VERIFICATION METHOD

Typically Test or Analysis (benchmark, load test, timing measurement).

6.1.6 CONSTRAINTS

Applicable constraints (target hardware, network conditions, power budget).

6.1.7 APPLICABLE STANDARDS

List of applicable standards, if any.

6.2 PE-02: [REQUIREMENT NAME]

Repeat the attribute block for each performance requirement.

7 PACKAGING & DEPLOYMENT REQUIREMENTS

Include a header paragraph specific to your product here. Packaging and deployment requirements identify how the delivered product will be packaged, distributed, and installed — how it will “look” when finished and delivered. Examples: software pre-loaded on hardware, delivered via signed installer or container image, customer-installable or not; hardware delivered assembled or as a kit; enclosure finish, labeling, and logos. Take care not to duplicate requirements found in other sections.

7.1 PK-01: [REQUIREMENT NAME]

7.1.1 DESCRIPTION

A detailed requirement description, e.g. *The application shall be delivered as a Docker image published to the customer’s private registry, deployable with a single documented command.*

7.1.2 RATIONALE

Why this packaging/deployment approach is required.

7.1.3 SOURCE

Source of the requirement.

7.1.4 PRIORITY

Critical / High / Moderate / Low / Future

7.1.5 VERIFICATION METHOD

Typically Demonstration or Inspection (perform a clean install/deploy and observe).

7.1.6 CONSTRAINTS

Applicable constraints (target environment, offline installation, media, size limits).

7.1.7 APPLICABLE STANDARDS

List of applicable standards, if any.

7.2 PK-02: [REQUIREMENT NAME]

Repeat the attribute block for each packaging/deployment requirement.

8 SAFETY REQUIREMENTS

Include a header paragraph specific to your product here. Safety requirements address hazards to people, property, and the environment arising from developing, operating, or maintaining the product: exposure to toxic materials, sharp edges, breakable glass, laser/IR eye exposure, electrical shock, stored mechanical or chemical energy, moving machinery, etc. The examples below are common to the CSE Senior Design laboratory; retain those that apply and add product-specific safety requirements.

8.1 SA-01: LABORATORY EQUIPMENT LOCKOUT/TAGOUT (LOTO) PROCEDURES

8.1.1 DESCRIPTION

Any fabrication equipment used in the development of the project shall be used in accordance with OSHA standard LOTO procedures. Locks and tags are installed on all equipment items that present use hazards, and only the course instructor or designated teaching assistants may remove a lock. All locks will be immediately replaced once the equipment is no longer in use.

8.1.2 RATIONALE

Prevents injury from unexpected energization or start-up of laboratory equipment during service or setup.

8.1.3 SOURCE

CSE Senior Design laboratory policy.

8.1.4 PRIORITY

Critical

8.1.5 VERIFICATION METHOD

Inspection (locks/tags present and controlled) and Demonstration (procedure followed).

8.1.6 CONSTRAINTS

Equipment usage, due to lock removal policies, is limited to the availability of the course instructor and designated teaching assistants.

8.1.7 APPLICABLE STANDARDS

OSHA 29 CFR 1910.147 — The control of hazardous energy (lockout/tagout).

8.2 SA-02: NATIONAL ELECTRIC CODE (NEC) WIRING COMPLIANCE

8.2.1 DESCRIPTION

Any electrical wiring shall be completed in compliance with all requirements specified in the National Electric Code, including wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

8.2.2 RATIONALE

Reduces risk of fire and electric shock in fabricated electrical assemblies.

8.2.3 SOURCE

CSE Senior Design laboratory policy.

8.2.4 PRIORITY

Critical

8.2.5 VERIFICATION METHOD

Inspection of wiring against NEC provisions.

8.2.6 CONSTRAINTS

High-voltage power sources, as defined in NFPA 70, will be avoided as much as possible to minimize hazards.

8.2.7 APPLICABLE STANDARDS

NFPA 70 (National Electrical Code).

8.3 SA-03: ROBOTIC MANIPULATOR SAFETY

8.3.1 DESCRIPTION

Robotic manipulators, if used, shall either be housed in a compliant lockout cell with all required safety interlocks, or be certified as a “collaborative” unit from the manufacturer.

8.3.2 RATIONALE

Protects personnel from injury by industrial robot motion.

8.3.3 SOURCE

CSE Senior Design laboratory policy.

8.3.4 PRIORITY

Critical

8.3.5 VERIFICATION METHOD

Inspection of interlocks/certification; Demonstration of safe operation.

8.3.6 CONSTRAINTS

Collaborative manipulators are preferred over non-collaborative units to minimize hazards. Sourcing and use of any required safety interlock mechanisms is the responsibility of the engineering team.

8.3.7 APPLICABLE STANDARDS

ANSI/RIA R15.06 — Industrial Robots and Robot Systems; RIA TR15.606 — Collaborative Robots.

9 SECURITY & PRIVACY REQUIREMENTS

Include a header paragraph specific to your product here. Security and privacy requirements address protection of information and system integrity: authentication and authorization, encryption in transit and at rest, key and secret management, input validation, auditing/logging, data retention and privacy, and secure update. Map these to the security characteristics of ISO/IEC 25010:2023 [2] (confidentiality, integrity, non-repudiation, accountability, authenticity) where useful.

9.1 SE-01: [REQUIREMENT NAME]

9.1.1 DESCRIPTION

A detailed, verifiable statement, e.g. *All network communication between the client and server shall be encrypted using TLS 1.3 or later; plaintext HTTP shall be refused.*

9.1.2 RATIONALE

The threat mitigated or the confidentiality/integrity/privacy obligation satisfied.

9.1.3 SOURCE

Source of the requirement (customer policy, regulation, standard, team).

9.1.4 PRIORITY

Critical / High / Moderate / Low / Future

9.1.5 VERIFICATION METHOD

Typically Test or Inspection (e.g. protocol scan, code review, penetration test).

9.1.6 CONSTRAINTS

Applicable constraints (regulatory data handling, export controls, key custody).

9.1.7 APPLICABLE STANDARDS

e.g. IETF RFC 8446 (TLS 1.3), NIST SP 800-63B, OWASP ASVS, FERPA/HIPAA/GDPR as applicable.

9.2 SE-02: [REQUIREMENT NAME]

Repeat the attribute block for each security/privacy requirement.

10 MAINTENANCE & SUPPORT REQUIREMENTS

Include a header paragraph specific to your product here. Maintenance and support requirements address the ongoing care of the product after delivery. Consider what a maintainer would need to correct defects and hardware failures “in the field”: who maintains the product and where, required troubleshooting manuals/guides, availability and documentation of source code, technical documentation for maintainers, and any unique tools or environments required. Relate these to the maintainability characteristic of ISO/IEC 25010:2023 [2] (modularity, reusability, analysability, modifiability, testability).

10.1 MS-01: [REQUIREMENT NAME]

10.1.1 DESCRIPTION

A detailed, verifiable statement, e.g. *The system shall provide structured logs sufficient to diagnose any unhandled error to the responsible subsystem without attaching a debugger.*

10.1.2 RATIONALE

Why this maintenance/support capability is required.

10.1.3 SOURCE

Source of the requirement.

10.1.4 PRIORITY

Critical / High / Moderate / Low / Future

10.1.5 VERIFICATION METHOD

Typically Inspection or Demonstration.

10.1.6 CONSTRAINTS

Applicable constraints (support staffing, access, documentation format).

10.1.7 APPLICABLE STANDARDS

List of applicable standards, if any.

10.2 MS-02: [REQUIREMENT NAME]

Repeat the attribute block for each maintenance/support requirement.

11 OTHER QUALITY ATTRIBUTES

Include a header paragraph specific to your product here. This section captures quality attributes and other requirements not covered above — for example reliability, availability, portability, compatibility, and adaptability. These map to product-quality characteristics of ISO/IEC 25010:2023 [2]. Also record here any requirement related to product architecture/design that the customer mandates, such as modularity, extensibility for future enhancements, or portability of the source across platforms (Windows, Linux, macOS, etc.).

11.1 QA-01: [REQUIREMENT NAME]

11.1.1 DESCRIPTION

A detailed, verifiable statement, e.g. *The service shall achieve at least 99.5% monthly availability measured at the public API endpoint.*

11.1.2 RATIONALE

Why this quality attribute matters to the customer.

11.1.3 SOURCE

Source of the requirement.

11.1.4 PRIORITY

Critical / High / Moderate / Low / Future

11.1.5 VERIFICATION METHOD

Inspection / Analysis / Demonstration / Test.

11.1.6 CONSTRAINTS

Applicable constraints.

11.1.7 APPLICABLE STANDARDS

List of applicable standards, if any.

11.2 QA-02: [REQUIREMENT NAME]

Repeat the attribute block for each additional quality requirement.

12 VERIFICATION & TRACEABILITY

ISO/IEC/IEEE 29148:2018 [1] requires that requirements be verifiable and bidirectionally traceable. This section collects the verification method for every requirement into a single matrix and records forward traceability to the design and test artifacts that will satisfy and confirm each requirement. Maintain this matrix as requirements evolve; it is the primary tool used during the Architectural Design, Detailed Design, and Test Plan phases to confirm coverage.

12.1 REQUIREMENTS VERIFICATION MATRIX

List every requirement identifier from Sections 3–11 with its declared verification method (Inspection, Analysis, Demonstration, Test) and the acceptance criterion that defines “pass.”

Req ID	Requirement (abbreviated)	Method	Acceptance criterion
FR-01	abbreviated requirement text	T	measurable pass condition
PE-01	abbreviated requirement text	T	measurable pass condition
SE-01	abbreviated requirement text	I	measurable pass condition

12.2 TRACEABILITY

Provide (or reference) a traceability matrix linking each requirement to (a) the architectural/detailed design element that realizes it and (b) the test case(s) that verify it. Early in the project the design and test columns may be marked “TBD” and completed as those documents are produced. Bidirectional traceability lets the team confirm that every requirement is designed and tested, and that no design element or test exists without a driving requirement.

Table 6: Requirements traceability (excerpt)

Req ID	Design element (ADS/DDS)	Test case (Test Plan)
FR-01	subsystem / component reference	TC-xx
PE-01	subsystem / component reference	TC-xx

13 FUTURE ITEMS

In this last section, reiterate all requirements assigned the **Future** priority. This is repetitive but necessary: it is a concise statement of features/functions that were considered and documented herein but will *not* be addressed in the prototype version of the product due to constraints of budget, time, skills, technology, or feasibility. Recording them preserves the rationale for a future release and keeps the traceability record complete. Use the same attribute format as the requirement sections above.

13.1 [REQUIREMENT NAME]

13.1.1 DESCRIPTION

Detailed requirement description ...

13.1.2 RATIONALE

Why the capability is desirable and why it is deferred.

13.1.3 SOURCE

Source of the requirement.

13.1.4 PRIORITY

Future

13.1.5 VERIFICATION METHOD

Inspection / Analysis / Demonstration / Test (as it would be verified if implemented).

13.1.6 CONSTRAINTS

The constraint(s) that make this infeasible for the current prototype.

13.1.7 APPLICABLE STANDARDS

List of applicable standards, if any.

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Life cycle processes — Requirements engineering,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2018.
- [2] ISO/IEC, “ISO/IEC 25010:2023 — Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.